

EE 446 Lab 7 - Data Collection for Magic Wand

Objective: Collect data for Lab 7 to be used in the development of a Magic Wand that can recognize digits drawn in the air.

Instructions:

1. Digits to Collect:
 - Each student must collect data for all digits from 0 to 9.
 - Number of Samples:
 - Collect 10 samples per digit, for a total of 100 data points. Ensure each digit is drawn in the air with a clear and steady motion. Remember, the quality of your data will directly impact the accuracy of the gesture recognition system.
2. Orientations: To ensure a robust dataset, collect data in different orientations. This includes varying the starting point, direction, and angle of the digit drawn. This will help the system generalize better to different user inputs.
3. IMPORTANT - Before You Begin Data Collection: Before accessing the data collection website, you must first upload the magic_wand_capture.ino sketch to your Arduino Nano 33 BLE Sense.
 - Download the ZIP file named magic_wand_capture.zip from Canvas →Files →Labs →Lab 7.
 - Extract it and open the .ino sketch inside the Arduino IDE.
 - Upload it to your Arduino board.
 - This step enables Bluetooth communication with the browser-based web tool.
 - Use Google Chrome when visiting the web tool, as Bluetooth pairing is only supported in Chrome.
4. Data Collection Tool:
 - After uploading the sketch, go to the following website to start collecting your digit data:
 - **Updated Link:** https://tinyml.seas.harvard.edu/magic_wand/
 - **Note:** Please ignore the link shown in the demo video if it differs-only use the updated link above.
5. Data Format:
 - Save each data point as a JSON file.
 - **Do not forget to label your drawings (In WebAPI you must also add labels for your strokes!!!)**
 - Ensure that the data is properly formatted and organized to facilitate easy processing.

6. Data Collection Demo:

- A demonstration video on how to collect data has been uploaded on Canvas → Files → Labs → Lab 7. Please watch the demo to understand the process and requirements clearly.

If you have any questions or issues during the data collection process, reach out to the instructor or TAs for assistance.

Note:

The success of the Magic Wand project depends on the quality and diversity of the data collected. Please follow the instructions carefully and collect your data as accurately as possible.

Happy Data Collecting!